

Finding Certifiable Bounds on Ground-State Properties Beyond Energy for Silicon

An Approach Using First-Principles Hamiltonians and Semi-Definite Programming

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Motivation: The Ground-State Problem

SDP & ground state energy problem

From Physics to a Computable Model

Application to Silicon

Conclusion & Outlook

Motivation: The Ground-State Problem

'HOLY GRAIL' of computational physics

Why Physicists Care:

- Quantum phases & phase transitions
- Emergent phenomena from correlations
- Quantum entanglement structure

Why Engineers Care:

- Battery & solar cell materials
- Superconductors & semiconductors
- Catalyst design & optimization

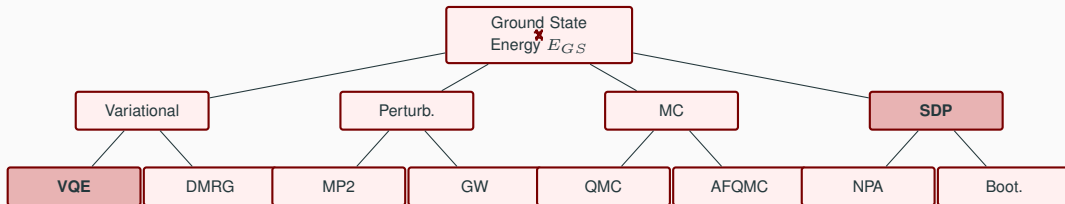
Key Equation

$$E_{\text{GS}} = \min_{|\psi\rangle \in \mathcal{H}} \langle \psi | H | \psi \rangle$$

The Core Challenge

$\dim(\mathcal{H}) = 2^N \Rightarrow$ Exponential complexity!

Ground State Energy Methods



Phasecraft Paper (Nature Commun. 2024)

"Towards near-term quantum simulation of materials"

Key Contributions:

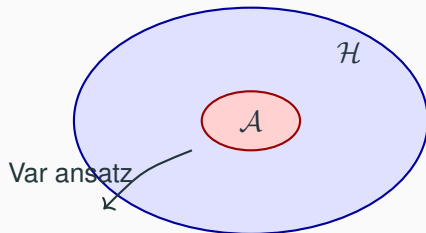
- Reduced VQE circuit depth for near-term quantum processors
- Constructed first-principles Hamiltonians for real materials
- Demonstrated feasibility with reduced qubit requirements

Our Goal:

- Use their Hamiltonian construction workflow
- Apply **Semi-Definite Programming** (SDP) as classical alternative
- Obtain rigorous **lower bounds** + properties beyond energy

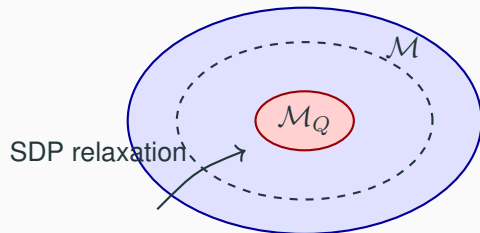
SDP & ground state energy problem

Variational vs SDP



Variational Methods:

- $\mathcal{A} \subset \mathcal{H}$ (inside approach)
- Choice of an Ansatz states less than physically possible states.
- Upper bounds on E_{GS}



SDP Methods:

- $\mathcal{M} \supset \mathcal{M}_Q$ (outside approach)
- Adds physically impossible elements, gets a simpler mathematical set.
- Lower bounds on E_{GS}

Ground State Energy and Quantum Moment Set

More precisely, we take

$$H = \sum_{i=1}^n \alpha_i h_i \quad \Rightarrow$$

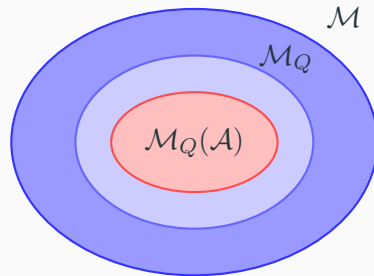
$$E_{\text{GS}} = \min_{\{\langle h_i \rangle\} \in \mathcal{M}_Q} \sum_{i=1}^n \alpha_i \langle h_i \rangle$$

Such that

$$\mathcal{M}_Q = \{ \{ \langle h_i \rangle \} : \exists |\psi\rangle \in \mathcal{H} \text{ such that } \langle h_i \rangle = \langle \psi | h_i | \psi \rangle, \forall i \},$$

In this sense, $\mathcal{M}_Q$ is the set of physically possible combinations of **quantum moments**.

$$(\langle h_1 \rangle, \langle h_2 \rangle, \dots, \langle h_n \rangle)$$



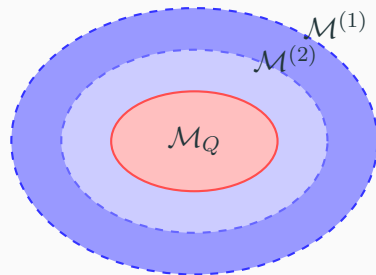
Mathematical Framework of SDP Hierarchy

The Moment Matrix:

- For operators $u, v \in \Gamma$ (set of monomials), define: $M_{uv} = \langle v^\dagger u \rangle$
- Quantum mechanics requires $M \succeq 0$.

Physical Constraints:

- Algebraic relations.
- Symmetries.
- Known bounds on observables.



$\mathcal{M}_Q$: True quantum moments

$\mathcal{M}^{(d)}$: SDP relaxation at level d

$$\mathcal{M}^{(1)} \supset \mathcal{M}^{(2)} \supset \dots \supset \mathcal{M}_Q$$

$$p^{(1)} \leq p^{(2)} \leq \dots \leq E_{\text{GS}}$$

What does a Semi-Definite Program look like?

$$p_{\min} = \min_{\{|\psi\rangle, X\}} \langle \psi | p(X) | \psi \rangle$$

subject to $g_j(X) \succeq 0, \quad j = 1, \dots, m_1$

$$\langle \psi | h_k(X) | \psi \rangle \geq 0, \quad k = 1, \dots, m_2$$

Ground state SDP

minimize $\langle H \rangle$

subject to $M \succeq 0$

Physical constraints

where p, g_j, h_k are polynomial functions of operator tuples $X = (X_1, \dots, X_n)$, and $\succeq 0$ denotes positive semidefiniteness.

Where M is the **quantum moment matrix**.

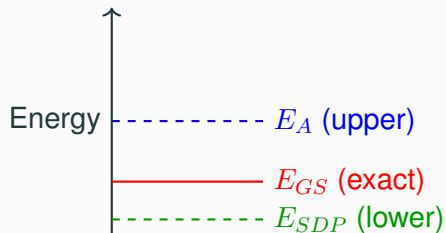
Advantages of SDP

- **Certifying bounds on E_{GS}**
upper + lower = trusted interval!
- **Going beyond Energy :**
We can extend the SDP formalism for any polynomial observable O :

$$o_{LB} = \min_{\{|\psi\rangle, X\}} \langle \psi | O(X) | \psi \rangle$$

subject to Physical constraints

$$E_A \leq \langle H \rangle \leq E_R$$



Proof of Concept: Graphene Tight-Binding Model

Before tackling Silicon, we validated our SDP implementation on the well-known graphene tight-binding model : $H = -t \sum_{\langle i,j \rangle, \sigma} (a_{i\sigma}^\dagger b_{j\sigma} + \text{h.c.})$

SDP Benchmark Results (4 unit cells)

Relaxation Level	Constraints Included	Solver Status	Lower Bound (eV/cell)
Level 1	Basic anti-commutation	Optimal	-22.40
Level 1	Full anti-commutation	Optimal	-19.60
Level 2	Basic anti-commutation	Feasible	-13.52
Level 2	Full anti-commutation	Optimal	-6.76
<i>Analytical Ground State Energy:</i>			<i>~-4.52</i>

Table 1: Implementation via 'ncpol2sdpa' + SDPA solver. Hopping parameter $t = 2.8$ eV.

From Physics to a Computable Model

The General Many-Body Hamiltonian

In second quantization, the Hamiltonian describing interacting electrons in a crystal is:

$$\hat{H} = \underbrace{\sum_{p,q,\sigma} t_{pq} c_{p\sigma}^\dagger c_{q\sigma}}_{\text{Kinetic + Potential Energy (Hopping)}} + \frac{1}{2} \underbrace{\sum_{p,q,r,s,\sigma\sigma'} V_{pqrs} c_{p\sigma}^\dagger c_{q\sigma'}^\dagger c_{r\sigma'} c_{s\sigma}}_{\text{Electron-Electron Interaction}}$$

The challenge lies in finding a suitable single-particle basis $\{\phi_p(\mathbf{r})\}$ to define the fermionic operators $c_{p\sigma}$ and compute the integral coefficients t_{pq} and V_{pqrs} .

- t_{pq} : Hopping matrix elements. $t_{pq} = \int d^3r \phi_p^*(\vec{r}) \left[-\frac{\hbar^2}{2m} \nabla^2 + U(\vec{r}) \right] \phi_q(\vec{r})$
- Coulomb tensor elements $V_{pqrs} = \frac{1}{2} \int d^3r d^3r' \phi_p^*(\vec{r}) \phi_q^*(\vec{r}') V(|\vec{r} - \vec{r}'|) \phi_r(\vec{r}') \phi_s(\vec{r})$

With $V(r) = q_e \frac{e^{-\mu r}}{4\pi\epsilon_0 r}$ is the Coulomb screened potential.

Why Standard Basis Choices Are Problematic

Bloch States $|\psi_{n\mathbf{k}}\rangle$

- Natural eigenstates of a perfect crystal.
- Completely delocalized over the entire solid.
- **Problem:** In this basis, the Coulomb tensor V_{pqrs} is densely populated.

Atomic Orbitals

- Chemically intuitive and localized.
- **Problem:** They are not the true eigenstates of the solid. Defining an accurate Hamiltonian from them is difficult and often empirical.

The Solution: A Bridge Between Physics and Chemistry

We need a basis that is both **derived from the true electronic structure** (like Bloch states) and **spatially localized** (like atomic orbitals).

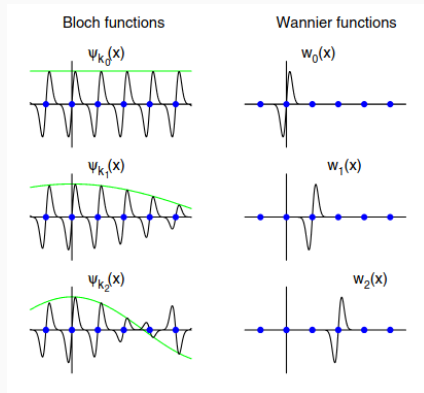
Maximally Localized Wannier Functions (MLWFs)

MLWFs provide this ideal basis. They are constructed by a unitary transformation of the Bloch states from a DFT calculation.

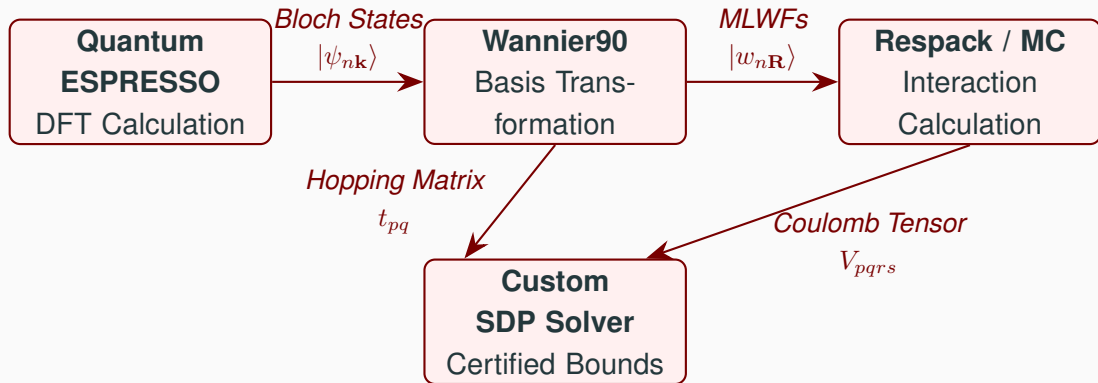
$$|w_{n\mathbf{R}}\rangle = \frac{V}{(2\pi)^3} \int_{\text{BZ}} e^{-i\mathbf{k}\cdot\mathbf{R}} \left(\sum_m U_{mn}^{(\mathbf{k})} |\psi_{m\mathbf{k}}\rangle \right) d\mathbf{k}$$

The matrices $U^{(\mathbf{k})}$ are chosen to minimize the total spatial spread of the resulting functions:

$$\Omega = \sum_n \left[\langle r^2 \rangle_{w_{n0}} - |\langle \mathbf{r} \rangle_{w_{n0}}|^2 \right]$$



The First-Principles Workflow



This pipeline transforms the complex quantum mechanical problem into a concrete, numerical Hamiltonian ready for advanced analysis.

Quick note on DFT

Core Idea: Replace the many-electron wavefunction with electron density $n(\mathbf{r})$ as the fundamental variable.

Kohn-Sham Practical Implementation

Maps interacting electrons to non-interacting particles in an effective potential:

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + U_{\text{eff}}(\mathbf{r}) \right] \phi_i(\mathbf{r}) = \varepsilon_i \phi_i(\mathbf{r})$$

$$U_{\text{eff}}(\mathbf{r}) = U_{\text{ext}} + V_{\text{H}}[n] + V_{\text{XC}}[n]$$

Strengths

- Ground state density & energy
- Practical for materials modeling
- Pseudopotentials reduce cost

Limitations

- $V_{\text{XC}}[n]$ must be approximated
- Band gap underestimation
- Challenging for strongly correlated

Application to Silicon

Silicon Workflow Implementation

Stage 1: Quantum ESPRESSO DFT

- Diamond structure (2 Si atoms), ONCV pseudopotentials
- **SCF**: 60 Ry cutoff, $8 \times 8 \times 8$ k-point mesh, PBE functional
- **NSCF**: $12 \times 12 \times 12$ k-point mesh for Wannierization
- **Successfully completed** $\rightarrow E_{GS}$, $n(r)$, *Set of bloch wave functions* $|\psi_{nk}\rangle$.

Stage 2: Wannier90 MLWFs

- 4 valence bands (sp^3 orbitals)
- Spread minimization: $\Omega \rightarrow \min$
- Final spread: $\sim 2.7 \text{ \AA}^2$ per WF
- **Successfully completed** $\rightarrow$ *Set of MLWF's* $|w_{n,\mathbf{R}}\rangle$, *Hopping matrix* t_{qp} .

Stage 3: Respack Coulomb Tensor

- **Technical compatibility issues**

Results: The Single-Particle Hamiltonian

The QE+Wannier90 pipeline successfully produced an accurate single-particle (tight-binding) Hamiltonian.

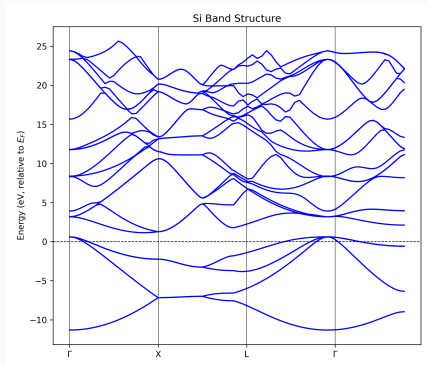


Figure 1: The electronic band structure of Silicon.

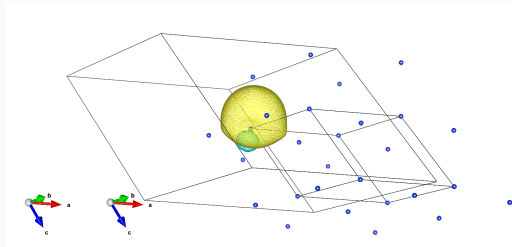
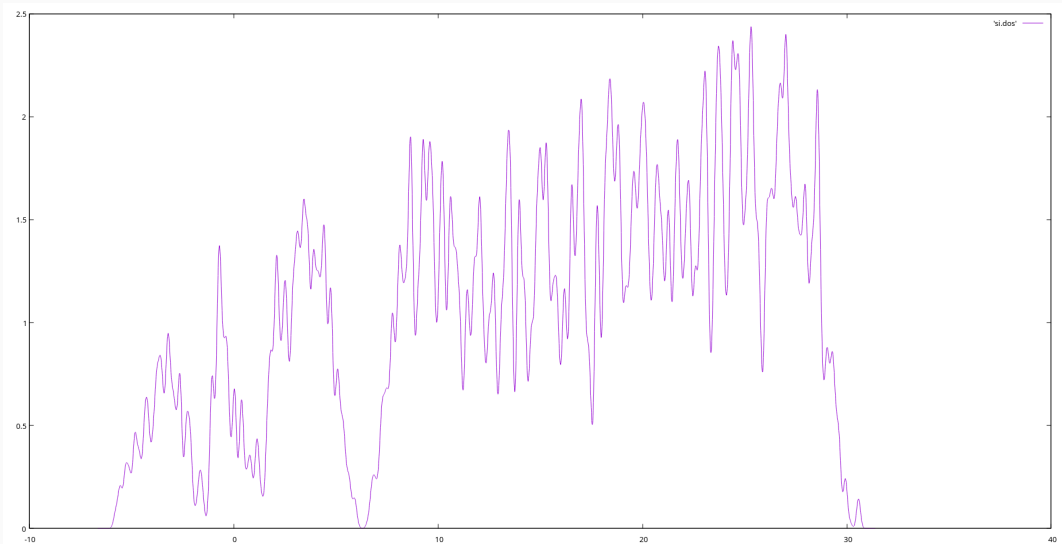


Figure 2: Visualization of an sp^3 -like MLWF, localized along a Si-Si bond. (Image from VESTA)

Density of states



The Bottleneck: Calculating Coulomb Integrals

Intended Goal:

- Calculate screened Coulomb interaction $W(\vec{r}, \vec{r}')$ and the screening length μ^{-1} .
- Generate full tensor V_{ijkl} in MLWF basis
- Complete many-body Hamiltonian: $\hat{H} = \hat{H}_0 + \hat{H}_{int}$
- use μ^{-1} to truncate space and use PBC.

Technical Roadblocks

- **Version incompatibility** between Wannier90 and RESPACK
- File format discrepancies in output data

⇒ **Impact:** Could not complete full workflow within internship timeframe

Monte Carlo Integration Alternative

Direct Calculation of Coulomb Tensor:

$$V_{ijkl} = \frac{1}{2} \int dr dr' w_i^*(r) w_j^*(r') V(|r - r'|) w_k(r') w_l(r)$$

MC Estimator:

$$V_{ijkl} \approx \frac{1}{N} \sum_{n=1}^N w_i^*(r_n) w_j^*(r'_n) V(|r_n - r'_n|) w_k(r'_n) w_l(r_n)$$

Advantages:

- Software-independent
- Progressive accuracy
- Focus on dominant terms
- Importance sampling possible

Implementation Strategy:

- Use MLWF numerical data
- Exploit exponential localization
- Apply Cauchy-Schwarz screening
- Parallelize over tensor indices

Future Work: Implementation underway

Conclusion & Outlook

Summary of Achievements

- Successfully implemented and validated a **Quantum ESPRESSO** → **Wannier90 pipeline** to derive a high-quality, localized Hamiltonian for Silicon from first principles.
- Developed and benchmarked a prototype **SDP framework** on a graphene model, demonstrating its capability to provide converging lower bounds on ground-state energy.
- Identified the primary technical blocker (Coulomb tensor calculation via Respack) and suggested a concrete and viable **alternative strategy** (Monte Carlo integration).
- Left a comprehensive documentation, a detailed Github repository for the pipeline and an interesting bibliography for any further research on the topic.

The groundwork from this project enables several exciting research directions:

- **Complete the Hamiltonian:** Implement the Monte Carlo integrator for V_{pqrs} to obtain the full interacting Hamiltonian for Silicon.
- **Run Full SDP Analysis:** Apply the SDP framework to the new Hamiltonian to compute the first certified lower bounds for ground-state properties of a realistic Silicon model.
- **Certify Beyond Energy:** Extend the method to find rigorous bounds on other key observables, such as charge correlations or entanglement measures.
- Exploring how physical constraints affect the accuracy of the SDP, and what would be the most beneficial ones to add.

Thank You!

Questions?

Code and data available at:

`https://github.com/Amddah-Mohamed/Si-Hamiltonian-for-SDP-.git`